

De Morgan's Laws

$$(A \cap B)^c = A^c \cup B^c$$

$$(A \cup B)^c = A^c \cap B^c$$

Partitions

$$A \cap A^c = \emptyset, \text{ whilst } A \cup A^c = S$$

$\Rightarrow A, A^c$ are collectively exhaustive

If we require both conditions for a group of sets, then those sets form a partition of S .

Definition: Consider sets $A_1, \dots, A_K$ where $\bigcup_{i=1}^K A_i = S$ and $A_i \cap A_j = \emptyset, i \neq j$

Then $A_1, \dots, A_K$ form a partition of S

$$\bigcup_{i=1}^K A_i = \sum_{i=1}^K |A_i| - \left\{ \bigcap_{i \in J} A_i : J \subseteq I, J \neq \emptyset \right\}$$